`knowledge\models\requirements-and-traceability.md`

Requirements and Traceability

Status: draft, revised from the owner's hand review of 2026-10-10 (decision D-020). The first draft (27 stakeholder requirements, 53 system requirements, written by Claude on 2026-10-10) was printed and marked up by the owner: `projects/nas-sos-capstone/handwritten/2026-10-10_requirements-and-traceability.pdf`. This file is that draft with the marks applied. Every requirement still has status `draft` in the model. Nothing is baselined. Not report text.

#TODO (owner) — what is left to review

- ☐ **(a) Read section 7.** It lists every mark that was a question, every mark Claude had to interpret, and every cut Claude made without a mark. Each line needs a yes or a correction.
- ☐ **(b) Confirm the three flight crew requirements you wrote** (SN-CRW-06, SN-CRW-07, SN-CRW-08) read as you meant, and the objective rows Claude attached to them.
- ☐ **(c) Decide the four weak ones.** SLR-PLN-05, SLR-PLN-08, SLR-FLW-03 and SLR-INF-07 were not marked, so they stayed, but their link to the study is thin. Keep or strike.
- ☐ **(d) Write the next tier for the thread** (three to five requirements) with the constituent system as the subject ("The ATC decision support system shall ..."). Your note on the old open item 2 reads "Turn them into HLRs". The starting material is the three statements SLR-INF-02 absorbed: gross weight, cost index and connection priority, each given to the party that decides. "The NAS shall" stays the subject at this tier.
- ☐ **(e) For each system requirement on the thread, name the verification case** (`cameo_models/nas_verification.sysml`) and give a measure and unit, or write "qualitative". On the thread: SLR-CLR-01, SLR-CLR-02, SLR-CLR-10, SLR-CLR-12, SLR-CLR-13, SLR-INF-01, SLR-INF-02, SLR-EXE-03, SLR-MEA-02.

The model is the source of truth (D-006). This file is a readable copy of two model files:

- `projects/nas-sos-capstone/cameo_models/requirements_stakeholders.sysml` holds the 12 stakeholder requirements.
- `projects/nas-sos-capstone/cameo_models/requirements_nas_system.sysml` holds the 28 system requirements and their allocation to systems.
- `projects/nas-sos-capstone/cameo_models/requirements_definitions.sysml` holds the attributes both use.

If this file and those disagree, the `.sysml` files win. Mark changes here by hand, then apply them to the `.sysml` files.

1. What is here

- **12 stakeholder requirements** in 5 stakeholder groups (was 27 in 8). Each says what that group wants from the NAS and names the objective rows it comes from in `knowledge/models/stakeholder-objective-ontology.md`.
- **28 system requirements** in 7 areas that follow the lifecycle of trajectory intent (was 53).
- **Every stakeholder requirement is answered by at least one system requirement, and every system requirement traces to a stakeholder requirement.** None is derived any longer.
- **Two of the 28 are goals, not requirements** (SLR-CLR-12, SLR-CLR-13). They are written with "should". Section 5.
- **Three describe what today's NAS does not do** (SLR-CLR-10, SLR-CLR-13, SLR-INF-02) and are tagged `improved` (D-018). They are what the experiment in `knowledge/models/experiment-definition.md` is about. The first draft had five; three of those five were merged into SLR-INF-02.
- Every system requirement is allocated to parts of `NationalAirspaceSystem` in `projects/nas-sos-capstone/cameo_models/nas_sysml_package_definitions.sysml`.

The trace runs: objective row, then stakeholder requirement, then system requirement, then constituent system.

The thread the project rests on: airline objective rows (fuel cost, schedule integrity, passenger connections) → SN-AIR-05 → SLR-INF-02 (the information) and SLR-CLR-13 (the goal it serves) → the `AirlinePerformanceShare` interface (D-019) → the verification cases → the experiment's outcome table.

2. The test each requirement had to pass

The owner's complaint about the first draft was that it described the NAS in general: it followed the detail in the model packages, and so held many true statements that have nothing to do with the project. The mark written most often on the print is "Not D-007".

D-007 (`decisions/decisions-log.md`) says an entity is modeled when it has "distinct decision authority or execution behavior dynamically coupled to in-scope decisions" and materially affects cost, safety, workload, schedule reliability or passenger value. The first draft used D-007 only to decide which stakeholder groups to include. The hand review applies the same test to each requirement, with "in-scope decisions" read as the decision the project studies:

> **How is a conflict between two aircraft in en-route cruise resolved, and what is the party that decides told about each aircraft?**

A requirement stays if it does one of three things:

1. It states what that decision must achieve or must not break (separation, a clearance the aircraft can fly, the airline's priorities).
2. It states what the party deciding knows, today or under the proposed change.
3. It fixes something the experiment takes as given and uses in its numbers (the flight plan, the fuel load, the cost function, the CO2 factor).

Requirements that are true of the NAS but do none of these were struck. Each kept requirement carries a note that begins "In the study:" and says which of the three it is. This reading of "Not D-007″ is Claude's statement of the owner's marks; it is recorded as D-020 for the owner to

confirm.

3. Stakeholder requirements (on the NAS)

The note under each table says why each requirement is in the study. It is the `rationale` attribute in the model.

Airline (Part 121 scheduled passenger operator)

D-007 treatment: Modeled system (Flight Operations).

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-AIR-01	The NAS shall provide a means for the airline to plan and fly each flight on the route, altitude and speed it chooses, so that the flight is flown ~~at its lowest operating cost~~ within safety and capacity limits.	Airline: Fuel cost; Airline: Delay cost	SLR-PLN-01, SLR-PLN-02, SLR-PLN-03, SLR-CLR-09, SLR-EXE-03, SLR-MEA-02	[B] `mori2022massCruise`, `munro2018managingVariability`
SN-AIR-02	The NAS shall provide a means for each flight to achieve arrival at its destination gate at its scheduled arrival time.	Airline: Schedule integrity; Airline: Aircraft utilization; Airline: Passenger connections	SLR-INF-07, SLR-APT-04	[C]
SN-AIR-03	The NAS shall inform the airline of airspace and airport constraints that will delay or reroute its flights early enough for the airline to re-plan.	Airline: Delay cost; Airline: Schedule integrity	SLR-FLW-03	As filed ensured to be separated. As uncertainty is realized a/c deviate & interventions needed. [A] `faa2025jo72103ee`
SN-AIR-05	The NAS shall take the airline's priorities for a flight (fuel against time, and protected connections) into account when a change to that flight's trajectory is required.	Airline: Fuel cost; Airline: Schedule integrity; Airline: Passenger connections	SLR-CLR-13, SLR-INF-02	[B] `schultz2012adaptiveClimb`, `coppenbarger1999climbPrediction`

- **SN-AIR-01**: In the study: the planned cruise altitude and speed are what a conflict resolution takes the aircraft away from, so this is the cost the experiment counts. Re-worded to the owner's pattern (provide a means for the airline to ... so that ...), hand review 2026-10-10.
- **SN-AIR-02**: In the study: the time side of the fuel-against-time trade, and the reason one flight's connections can outweigh its fuel (case D of the experiment). Owner's wording, hand review 2026-10-10.
- **SN-AIR-03**: In the study: context only. Constraints enter the scenario as givens; the study does not examine how they are set. Owner's wording, hand review 2026-10-10.
- **SN-AIR-05**: In the study: this is the requirement the experiment is about (knowledge/models/experiment-definition.md section 2). It is not met today: the controller does not receive weight or cost index. Owner's wording, hand review 2026-10-10.

Air traffic control and traffic flow management (FAA Air Traffic Organization)

D-007 treatment: Modeled system (Airspace Management).

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-ATC-01	The NAS shall keep every controlled aircraft separated from every other aircraft by at least the applicable separation minimum.	ATC / ANSP: Safety/separation	SLR-PLN-06, SLR-CLR-01, SLR-CLR-02, SLR-CLR-12, SLR-INF-01, SLR-EXE-02	[A] `faa2025jo711065bb`
SN-ATC-02	The NAS shall keep the number of aircraft in each sector and at each airport within its declared capacity.	ATC / ANSP: Capacity; ATC / ANSP: Sector workload; ATC / ANSP: Traffic complexity	SLR-PLN-06, SLR-PLN-08, SLR-APT-02	[A] `faa2025jo72103ee`

- **SN-ATC-01**: In the study: the hard constraint every resolution option must meet. It is never traded against cost (knowledge/models/experiment-definition.md section 6).
- **SN-ATC-02**: In the study: context. It is the capacity limit named in SN-AIR-01, and it carries the controller workload rows now that SN-ATC-04 is struck. Not marked in the hand review.

Flight crew (captain and first officer)

D-007 treatment: Modeled system (Flight Crew).

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-CRW-03	The NAS shall give the flight crew only trajectory instructions that the aircraft can fly, with enough time to set them up.	Flight Crew: Workload; Flight Crew: Safety	SLR-CLR-10, SLR-INF-02, SLR-EXE-03, SLR-EXE-06	[SME]
SN-CRW-06	The NAS shall provide the flight crew clearances that account for air traffic.	Flight Crew: Safety	SLR-CLR-01, SLR-CLR-02	[SME]
SN-CRW-07	The NAS shall inform the flight crew of weather, traffic and traffic management initiatives.	Flight Crew: Safety; Flight Crew: Procedural compliance	SLR-PLN-07, SLR-FLW-03, SLR-INF-06	[SME]

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-CRW-08	The NAS shall facilitate control of the aircraft across regional sectors, coordinating controls, clearances and handoffs.	Flight Crew: Workload; Flight Crew: Procedural compliance	SLR-CLR-04	[SME]

- **SN-CRW-03**: In the study: a climb the aircraft is too heavy to make is the clearance the baseline can issue and the improved architecture avoids. The only flight crew requirement of the first draft the owner kept.
- **SN-CRW-06**: Written by the owner in the hand review of 2026-10-10 (flight crew note 1). In the study: the clearance that resolves the conflict is the thing the crew receives. The objective row was chosen by Claude; owner to confirm.
- **SN-CRW-07**: Written by the owner in the hand review of 2026-10-10 (flight crew note 2). In the study: what the crew knows when it accepts, refuses or asks to change a clearance. The objective rows were chosen by Claude; owner to confirm.
- **SN-CRW-08**: Written by the owner in the hand review of 2026-10-10 (flight crew note 3). In the study: the conflict is resolved by one sector controller, whose authority ends at the sector boundary. The objective rows were chosen by Claude; owner to confirm.

Regulator (FAA as rule maker; Department of Transportation)

D-007 treatment: Boundary actor.

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-GOV-01	The NAS shall conduct every flight within the operating rules of 14 CFR and the FAA’s air traffic orders.	none (see note)	SLR-PLN-04, SLR-PLN-05, SLR-CLR-08, SLR-EXE-06, SLR-MEA-01	[A] `faa2025jo711065bb`, `faa2025jo72103ee`, 14 CFR 91, 14 CFR 121

- **SN-GOV-01**: In the study: the rules are held fixed (D-007), so both the baseline and the improved architecture must stay inside them. Governance has no table in knowledge/models/stakeholder-objective-ontology.md; the objective is ‘Regulatory Compliance’ in the enterprise objective hierarchy in knowledge/models/stakeholder-register.md. The owner judged adding a Governance table not important (hand review 2026-10-10).

Communities and environmental organizations

D-007 treatment: Boundary actor (acts through Governance per D-007).

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-ENV-01	The NAS shall keep the CO2 emitted by each flight as low as safety, economics and capacity allow.	Environmental / Societal: CO2	SLR-PLN-03, SLR-CLR-13, SLR-MEA-02	[C] `icao2024icecMethodology`

- **SN-ENV-01**: In the study: CO2 is fuel burned times 3.16 (D-015), so the experiment’s fuel column is also this stakeholder’s column. ‘Economics’ was added by the owner with a question mark (hand review 2026-10-10); owner to confirm.

Groups with no requirement after the review: Airport operator, Passengers, Military airspace users. See section 6 for what was struck and section 8 for the objective rows left without a requirement.

4. System requirements

Columns: **Traces to** is the stakeholder requirement it answers. **Allocated to** names the constituent systems, the accountable one first. **Check** is how it would be verified: inspection of the model, analysis (calculation or simulation), or demonstration (a scenario walked through on the activity diagram). **Measure** is filled only where there is something to measure; a number appears only where a source gives it. **State** is `improved` where today’s NAS does not do it (D-018), otherwise `baseline`. A statement written with “should” is a goal (section 5). The note under each table says why each one is in the study and records the owner’s mark.

Flight planning and release

Place on the trajectory-intent chain: mission plan -> flight plan.

ID	Requirement	Kind	State	Traces to	Allocated to	Check	Measure	Basis
SLR-PLN-01	The NAS shall produce for each scheduled flight a flight plan that states its route, cruise altitude, cruise speed and fuel load.	Functional	baseline	SN-AIR-01	Dispatcher	Inspection		[B] `berry2011aocActors`
SLR-PLN-02	The NAS shall compute each flight plan from the performance of the aircraft assigned to the flight.	Functional	baseline	SN-AIR-01	AOC decision support (flight planning system); Load planner	Analysis		[B] `mori2022massCruise`
SLR-PLN-03	The NAS shall select each flight plan’s route, altitude and speed to	Performance	baseline	SN-AIR-01,	AOC decision support (flight	Analysis	flight cost = fuel cost + cost index x flight	[B] `mori2022massCruise`

Can do multiple HLRs, which reflect different customers / instantiations: lo-fi, hi-fi, & rules of thumb per type.

This is an airline's perogative. Maximize revenue, profit, or min cost.

ID	Requirement	Kind	State	Traces to	Allocated to	Check	Measure	Basis
	minimize the airline's stated cost of fuel and flight time.			SN-ENV-01	planning system)		time (USD per flight)	
SLR-PLN-04	The NAS shall load each flight with fuel to reach its destination, then its most distant alternate, then fly 45 minutes at normal cruise.	Constraint	baseline	SN-GOV-01	Dispatcher	Inspection	fuel remaining after destination and alternate (minutes at normal cruise). Value: 45	[C] 14 CFR 121.639
SLR-PLN-05	The NAS shall release a flight only when the dispatcher and the captain both agree that it can be flown safely as planned.	Functional	baseline	SN-GOV-01	Dispatcher; Flight deck crew	Inspection		[A] 14 CFR 121.533
SLR-PLN-06	The NAS shall file each flight plan with air traffic control before the flight departs.	Interface	baseline	SN-ATC-01, SN-ATC-02	Dispatcher; Air traffic control	Inspection		[A] `faa2025aim`, `faa2025jo72103ee`
SLR-PLN-07	The NAS shall deliver the dispatch release, flight plan, weather, notices to airmen (NOTAMs) and load sheet to the flight crew before departure.	Interface	baseline	SN-CRW-07	Dispatcher; Load planner	Inspection		[B] `seamster2011collabSystems`
SLR-PLN-08	The NAS shall plan each flight to comply with the traffic management initiatives published for its time of flight.	Constraint	baseline	SN-ATC-02	Dispatcher; AOC ATC coordinator	Inspection		[A] `faa2025jo72103ee`

- **SLR-PLN-01**: In the study: the flight plan sets the cruise altitude, speed and fuel load each aircraft has when the conflict occurs.
- **SLR-PLN-02**: The owner struck 'the planned take-off weight' (hand review 2026-10-10). In the study: the cost-optimal altitude depends on mass (mori2022massCruise, Figure 1), which is why two aircraft of one type can want different levels.
- **SLR-PLN-03**: In the study: this cost (fuel plus cost index times time) is the one the experiment scores each resolution option with. The owner asked whether 'select' should read 'provide clearance for'. Left as 'select': the airline's flight planning system is inside the NAS and makes the selection; ATC's part is covered by SLR-PLN-06 and SLR-CLR-09. Owner to confirm.
- **SLR-PLN-04**: The owner asked whether this is relevant for the weight assumptions. It is: the fuel load is part of the take-off weight, and the reserve sets the least fuel, and so the least weight, an aircraft can have in cruise. Number and paragraph are from general knowledge; confirm against 14 CFR 121.639 before baselining.
- **SLR-PLN-05**: In the study: a weak link. It fixes who agreed the fuel and route the aircraft carries into cruise. Not marked in the hand review; a candidate to strike.
- **SLR-PLN-06**: In the study: the filed plan is the only statement of the airline's intent the controller has today, so it is the baseline that the proposed data share adds to. No lead time is set; the summary of faa2025aim records a 45 to 46 minute point before departure after which a filed plan is locked.
- **SLR-PLN-07**: In the study: the load sheet is how the actual weight reaches the crew and the flight management system.
- **SLR-PLN-08**: In the study: context. Initiatives are givens in the scenario. 'Special use airspace activations' was dropped from the wording when the military requirement was struck. Not otherwise marked in the hand review; a candidate to strike.

Traffic flow management

Place on the trajectory-intent chain: flight plan -> ATC constraints (before departure and strategic).

ID	Requirement	Kind	State	Traces to	Allocated to	Check	Measure	Basis
SLR-FLW-03	The NAS shall notify each affected airline of a traffic management initiative before its affected flights depart.	Interface	baseline	SN-AIR-03, SN-CRW-07	Command Center (ATCSCC); AOC ATC coordinator	Inspection		[A] `faa2025jo72103ee`, `berry2011aocActors`

- **SLR-FLW-03**: In the study: context. The only traffic flow management requirement kept, because SN-AIR-03 and SN-CRW-07 both name being informed of initiatives. The owner's note on this section: initiatives are not what the study examines.

Clearance, separation and tactical trajectory change

Place on the trajectory-intent chain: ATC constraints -> trajectory negotiation.

ID	Requirement	Kind	State	Traces to	Allocated to	Check	Measure	Basis
SLR-CLR-01	The NAS shall keep each pair of controlled aircraft separated by at least the minimum that JO 7110.65BB sets for the airspace they are in.	Performance	baseline	SN-ATC-01, SN-CRW-06	Air traffic control; En-route center (ARTCC); Terminal approach control (TRACON)	Analysis	lateral and vertical distance between each pair of aircraft (NM and ft)	[A] `faa2025jo711065bb`
SLR-CLR-02	The NAS shall predict a loss of separation between controlled aircraft before it occurs.	Functional	baseline	SN-ATC-01, SN-CRW-06	ATC decision support	Analysis		[B] `engility2014tasarAlaska`
SLR-CLR-04	The NAS shall provide a two-way communication path from the flight crew to the controlling facility and to the dispatcher for the whole [redacted] flight.	Interface	baseline	SN-CRW-08	Aircraft communication system; Air traffic control; Dispatcher	Inspection		[SME]
SLR-CLR-08	The NAS shall allow the captain to deviate from a clearance when safety requires it, with air traffic control told as soon as practical.	Functional	baseline	SN-GOV-01	Flight deck crew	Inspection		[A] 14 CFR 91.3
SLR-CLR-09	The NAS shall answer each flight crew request for a route, altitude or speed change with an approval, an amended clearance or a refusal.	Functional	baseline	SN-AIR-01	Air traffic control; Flight deck crew	Demonstration		[B] `engility2014tasarAlaska`
SLR-CLR-10	The NAS shall issue only clearances that the aircraft can fly.	Constraint	improved	SN-CRW-03	Air traffic control	Analysis		[B] `coppenbarger1999climbPrediction`, `engility2014tasarAlaska`
SLR-CLR-12	The NAS should resolve each predicted conflict with the fewest control instructions that restore separation.	Performance (goal)	baseline	SN-ATC-01	Air traffic control	Analysis	instructions issued plus follow-up instructions, per conflict (count)	[B] `rantanen2012conflictManeuvers`, `kirwan2001coraStrategies` Why? This isn't a safety objective. If not overworked, controllers can give as many as they want!
SLR-CLR-13	The NAS should choose, among the maneuvers that restore separation ~~with the same~~	Performance (goal)	improved	SN-AIR-05, SN-ENV-01	ATC decision support; Air traffic control	Analysis	sum over the aircraft involved of fuel cost plus cost index x	[B] `schultz2012adaptiveClimb`, `mori2022massCruise`, `kirwan2001coraStrategies`

ID	Requirement	Kind	State	Traces to	Allocated to	Check	Measure	Basis
	number of control instructions, the one with the lowest combined cost to the aircraft involved.						time (USD per conflict)	

- **SLR-CLR-01**: In the study: the hard constraint every resolution option must meet; it is not traded (knowledge/models/experiment-definition.md section 6). For en-route radar the minimum is commonly 5 NM or 1,000 ft; those numbers are from general knowledge and are left out of the threshold until the paragraph is confirmed.
- **SLR-CLR-02**: In the study: the conflict has to be predicted before any option can be chosen. No look-ahead time is set; engility2014tasarAlaska page 9 uses an eight-minute probe in its own model, which is a candidate.
- **SLR-CLR-04**: ‘Dispatcher too’ was added by the owner (hand review 2026-10-10). In the study: the clearance, the crew’s ‘unable’ and any crew request cross this path. It was a derived requirement in the first draft; it now answers SN-CRW-08.
- **SLR-CLR-08**: 14 CFR 91.3. In the study: the owner asks whether leaving the choice to the crew, with a response delay, should be one of the experiment’s options. Open.
- **SLR-CLR-09**: The owner asked what its relevance is. In the study: a crew request is the only way today that what the aircraft knows about its own weight and best altitude reaches the controller, so it is the baseline counterpart of the proposed data share. Strike it if that comparison is not wanted.
- **SLR-CLR-10**: The owner struck ‘at its current weight’: weight is not required, but it helps when it is included in the model behind a decision aid. Not fully met today: the ground system assumes one nominal weight per aircraft type (coppenbarger1999climbPrediction), and the crew answers ‘unable’ when a clearance cannot be flown. The rule in D-013 (offer a climb only at or below FL350) stands in for the missing knowledge. Kept as improved (D-018) because that fact has not changed; owner to confirm.
- **SLR-CLR-12**: A goal, not a requirement (owner, hand review 2026-10-10): it is not required, but it is probably the best way to achieve and maintain separation and preserve safety. In the study: the experiment’s workload column (instructions issued plus follow-ups).
- **SLR-CLR-13**: A goal, not a requirement (owner, hand review 2026-10-10). The held sources agree: the controller’s aim is to separate aircraft ‘first safely, and then expeditiously’ (kirwan2001coraStrategies, p. 4), and no held source read so far makes lowest cost to the aircraft a duty. Not pursued today, because the deciding party lacks the information in SLR-INF-02. In the study: the experiment measures what pursuing it is worth.

Information for trajectory decisions

Place on the trajectory-intent chain: what the party making each decision must be given.

ID	Requirement	Kind	State	Traces to	Allocated to	Check	Measure	Basis
SLR-INF-01	The NAS shall give the sector controller the filed route, requested altitude and requested speed of each aircraft before it enters the sector.	Interface	baseline	SN-ATC-01	Air traffic control; En-route center (ARTCC)	Inspection		[A] `faa2025jo711065bb`
SLR-INF-02	The NAS shall give the party that decides a change to an aircraft’s trajectory the information needed to preserve safety and pursue optimality.	Interface	improved	SN-AIR-05, SN-CRW-03	Flight management system; Airline operations center; ATC decision support; Air traffic control	Analysis		[B] `schultz2012adaptiveClimb`, `coppenbarger1999climbPrediction`, `sheth2010creditsConcept`
SLR-INF-06	The NAS shall give the dispatcher and the flight crew the reported and forecast weather, including turbulence, along the planned route.	Interface	baseline	SN-CRW-07	Airline operations center; Flight deck crew	Inspection		[C] `seamster2011collabSystems`
SLR-INF-07	The NAS shall send the airline and the destination airport an updated arrival estimate whenever a flight’s estimate changes by more than 5 minutes.	Interface	baseline	SN-AIR-02	Flight management system; Flight deck crew; Airline operations center; Airport management system	Inspection	change in estimated arrival time that triggers an update (minutes). Value: 5	[B] `seamster2011collabSystems`

- **SLR-INF-01**: In the study: this is what the controller knows about each aircraft today (case A of the experiment).
- **SLR-INF-02**: Owner's wording (hand review 2026-10-10). It replaces three statements that named gross weight, cost index and connection priority one at a time (the first draft's SLR-INF-02, SLR-INF-03 and SLR-INF-04); the owner judged them implied. Those three items are the attributes of AirlinePerformanceShare in nas_sysml_package_definitions.sysml (D-019) and are where the next tier of requirements starts. Not met today: the controller does not receive weight or cost index (schultz2012adaptiveClimb, page 2). Cases C and D of the experiment.
- **SLR-INF-06**: 'Weather' was added by the owner (hand review 2026-10-10). In the study: wind and turbulence are reasons a level is wanted or refused apart from weight.
- **SLR-INF-07**: In the study: how the airline learns what a trajectory change did to the arrival time. The 5 minutes is the crew-to-dispatch practice in seamster2011collabSystems (survey item: notify dispatch of ETA changes greater than 5 minutes). Extending the same trigger to the airport is this draft's proposal, not the source's.

Onboard execution

Place on the trajectory-intent chain: FMS intent -> guidance commands -> aircraft motion.

ID	Requirement	Kind	State	Traces to	Allocated to	Check	Measure	Basis
SLR-EXE-02	The NAS shall fly each aircraft along its cleared route and at its cleared altitude within the navigation and altitude tolerances of the airspace.	Performance	baseline	SN-ATC-01	Auto flight control system; Flight management system	Analysis	cross-track distance from the cleared route; difference from the cleared altitude (NM and ft)	[SME]
SLR-EXE-03	The NAS shall predict ~~on board~~ each flight's arrival time, fuel at destination, optimum altitude and maximum altitude from the aircraft's estimated performance.	Functional	baseline	SN-AIR-01, SN-CRW-03	Flight management system	Inspection		[SME] `mori2022massCruise`
SLR-EXE-06	The NAS shall alert the flight crew when the fuel predicted at the destination falls below the required reserve.	Functional	baseline	SN-GOV-01, SN-CRW-03	Flight management system	Inspection		[SME]

- **SLR-EXE-02**: Owner (hand review 2026-10-10): this has to be accounted for in the separation requirements. In the study: how far an aircraft may stray from its clearance is part of the margin a resolution must leave. No tolerance is set; it depends on the navigation specification of the route, which the scenario has not chosen.
- **SLR-EXE-03**: 'Estimated performance' is the owner's wording; it replaces 'actual weight, the forecast wind and the cost index'. This is the item FmsPerformancePrediction in nas_sysml_package_definitions.sysml. In the study: it is the aircraft's own knowledge of its optimum and maximum altitude, which the improved architecture would share.
- **SLR-EXE-06**: The owner left two questions (hand review 2026-10-10): re-word it as 'provide no clearance that results in fuel at the destination below the threshold', or move it down a tier as a requirement on the crew? Left unchanged until answered.

Airport and surface

Place on the trajectory-intent chain: start and end of the chain on the ground.

ID	Requirement	Kind	State	Traces to	Allocated to	Check	Measure	Basis
SLR-APT-02	The NAS shall sequence arrivals and departures so that each runway is used at its declared capacity whenever demand is at or above that capacity.	Performance	baseline	SN-ATC-02	Terminal approach control (TRACON); Control tower	Analysis	runway operations per hour against the declared capacity (operations per hour)	[A] `faa2025jo72103ee`
SLR-APT-04	The NAS shall have each aircraft fuelled, loaded and sealed by its scheduled off-block time.	Performance	baseline	SN-AIR-02	Ground handling; Fuel distribution system; Flight deck crew	Demonstration	actual off-block time minus scheduled off-block time (minutes)	[B] `schultz2017turnaround`

- **SLR-APT-02**: 'Capacity' replaces 'rate' (owner, hand review 2026-10-10). Owner: not necessary to the study, but it informs prioritization decisions.
- **SLR-APT-04**: Owner (hand review 2026-10-10): this speaks to the unknowns in weights and to when the information is available. The final weight is known only when loading ends. 'Sealed' replaces 'closed up' and carries the owner's question mark.

Measurement and records

Place on the trajectory-intent chain: back up the chain: what is reported to the authorities and stakeholders.

ID	Requirement	Kind	State	Traces to	Allocated to	Check	Measure	Basis
SLR-MEA-01	The NAS shall record each change to an aircraft's trajectory, to its clearance, or to the constraints imposed on it, with the authority that issued it and the time.	Functional	baseline	SN-GOV-01	Dispatcher; Air traffic control; Command Center (ATCSCC)	Inspection		[A] `faa2025jo72103ee`
SLR-MEA-02	The NAS shall report the fuel burned and the CO2 emitted by each flight, at 3.16 kg of CO2 per kg of fuel.	Performance	baseline	SN-ENV-01, SN-AIR-01	Flight management system; Airline operations center	Analysis	CO2 emitted per kg of fuel burned (kg CO2 per kg fuel). Value: 3.16	[A] `icao2024icecMethodology`

- **SLR-MEA-01**: Re-worded from the owner's note (hand review 2026-10-10), which was written as a question; owner to confirm. In the study: the record is what lets a resolution be scored after the fact.
- **SLR-MEA-02**: In the study: the experiment's fuel column, and the CO2 figure derived from it. icao2024icecMethodology page 6, accepted in D-015. The number is a conversion factor, not a limit.

5. Goals

A goal is a statement the owner judged "not required". It stays in the model because the experiment measures it, but it is written with "should" and carries `obligation = goal` (new attribute in `requirements_definitions.sysml`, D-020). A goal is measured; it is not passed or failed.

- **SLR-CLR-12** (fewest control instructions per conflict). Owner's note: "This isn't required... probably the 'best' way to achieve & maintain separation & preserve safety." It is the experiment's workload column.
- **SLR-CLR-13** (lowest combined cost among maneuvers with the same instruction count). Owner's note: "Not required, but a goal. What do sources say?" The held sources agree with the owner. Kirwan and Flynn record the controller's aim as to separate aircraft "first safely, and then expeditiously" (`kirwan2001coraStrategies`, p. 4, as quoted in `evidence/literature-notes/summaries/4 - kirwan2001coraStrategies.md`). The summaries of JO 7110.65BB and JO 7210.3EE held here record no duty to pick the cheapest maneuver for the aircraft. The summaries were searched for this; the orders themselves were not re-read. Schultz and the TASAR study treat lower airline cost as a benefit to be gained, not as a rule in force.

6. What was struck, and the owner's reason

The full text of each struck requirement is in the marked-up print, `projects/nas-sos-capstone/handwritten/2026-10-10_requirements-and-traceability.pdf`. The first draft was never committed to git, so that PDF is the record of it. "Owner's mark" quotes the handwriting. Where the column says "no mark", the cut is Claude's, made because the requirement's only purpose went with something the owner did strike; section 7 lists these for confirmation.

Stakeholder requirements struck (18 of 27)

ID	It said, in short	Owner's mark
SN-AIR-04	The airline chooses which of its own flights takes an assigned delay.	"Fact check. This is about slot swapping, I think... seems like it's only on strategic timelines, not tactical like I'm studying." Checked: it is. See section 7.
SN-AIR-06	Keep the airline's commercially sensitive flight data private.	"Important & real, but not pertinent to my project per D-007!" It becomes an assumption (D-020 item 4).
SN-ATC-03	The path each aircraft will fly is known to its controller in advance.	"Not D-007"
SN-ATC-04	Control actions per sector stay within what one sector team can sustain.	"Not D-007"
SN-ATC-05	Unavoidable delay is shared equitably among airspace users.	no mark. Its only system requirements were about traffic management initiatives.
SN-APT-01, SN-APT-02, SN-APT-03	Reliable times for gates; declared runway capacity used; short taxi queues.	"All of these are not D-007"
SN-CRW-01	The captain's final authority is preserved.	crossed out
SN-CRW-02	The crew has the information for each phase before it begins.	crossed out; replaced by the owner's SN-CRW-07
SN-CRW-04	The crew can ask for a trajectory change and get an answer.	crossed out
SN-CRW-05	Flight and duty time limits are kept.	crossed out, "Not D-007"
SN-PAX-01, SN-PAX-02, SN-PAX-03	No injury; itinerary completed on time; known turbulence avoided.	crossed out, "Not D-007"
SN-GOV-02	One identified authority is accountable for each trajectory decision.	crossed out. "IDK about this one... the Note is an exception!"

ID	It said, in short	Owner’s mark
SN-ENV-02	Noise near airports within published procedures.	crossed out, “Not important for cruise.”
SN-MIL-01	Military operations have sole use of active special use airspace.	crossed out

System requirements struck (25 of 53)

ID	It said, in short	Owner’s mark
SLR-PLN-09	Crew duty period within 14 CFR Part 117.	crossed out, “Not important to project.”
SLR-FLW-01, SLR-FLW-02, SLR-FLW-04, SLR-FLW-05	Forecast demand against capacity; issue an initiative; slot substitution; assign delay equitably.	no mark on the rows. On the section: “TMIs aren’t what I’m studying. Should they be? Miles/Minutes in trail sub-optimal... I don’t think I want to.”
SLR-FLW-06, SLR-FLW-07	Depart within the EDCT window; publish the special use airspace schedule.	crossed out
SLR-CLR-03	Position, altitude and identity of each aircraft to the controlling facility.	crossed out, “of course. Not relevant to project.”
SLR-CLR-05	A cleared trajectory changes only by a clearance from the controlling facility.	crossed out, “Not relevant to project.”
SLR-CLR-06	The crew reads back each clearance.	crossed out, “Not relevant”
SLR-CLR-07	Control is transferred only when the receiving controller accepts.	crossed out, “Not relevant”
SLR-CLR-11	Holding instructions at least 5 minutes ahead.	crossed out, “Not Relevant!”
SLR-CLR-14	Civil aircraft kept out of active special use airspace.	crossed out, “Not Relevant.”
SLR-INF-03, SLR-INF-04	Cost index, and connection priority, to the party that decides.	crossed out. Merged into SLR-INF-02: “Implied. Why add separately?”
SLR-INF-05	Shared airline data disclosed only to the deciding parties.	crossed out, “Maybe not needed if exposed as API.”
SLR-INF-08	One flight identifier and one time reference on every exchange.	crossed out, “Sure, but why relevant for my project?”
SLR-EXE-01	The cleared path is held in the flight management system.	crossed out, “why needed?”
SLR-EXE-04	The flown trajectory changes only through flight deck controls.	crossed out, “of course. Why needed?”
SLR-EXE-05	Onboard position determined to the accuracy the route requires.	crossed out, “Not needed.”
SLR-APT-01	A gate is assigned before landing.	crossed out, “Not relevant”
SLR-APT-03	A known departure delay is absorbed at the gate.	crossed out, “I didn’t list this, as the SME. Ground delay isn’t my focus!”
SLR-APT-05	Noise abatement procedures are assigned.	crossed out, “Not needed.”
SLR-MEA-03	Gate arrival time reported against schedule.	crossed out, “Relevance?”
SLR-MEA-04	Delay from initiatives reported by airline.	no mark. It existed only so SLR-FLW-05 could be checked.

On the whole of the old section 5 (derived requirements) the owner wrote: “This has nothing to do w/ my project & doesn’t meet D-007!” Four of the five derived requirements are struck above. The fifth, SLR-CLR-04 (the communication path), was not crossed out on its own row and carries the owner’s addition “Dispatcher too”; it is kept and now answers SN-CRW-08.

7. Questions on the print, readings Claude had to make, and cuts without a mark

Questions the owner wrote, and the answer given here.

1. **SN-AIR-04, “Fact check... slot swapping... only on strategic timelines?”** Correct. `evidence/literature-notes/summaries/5 - faa2025jo72103ee.md` records JO 7210.3EE paragraph 18-10-12 as the user options inside a ground delay program, which operators coordinate with the Command Center before departure. It does not touch a conflict in cruise. Struck.
2. **Traffic flow management, “TMIs aren’t what I’m studying. Should they be?”** No, on the evidence held. The experiment is one sector controller and one crossing pair; an initiative reaches it only as a given. Studying initiatives would be a second project (the Command Center’s allocation rules, CTOP, slot substitution). One requirement was kept, SLR-FLW-03, because your own SN-AIR-03 and SN-CRW-07 both say “inform ... of TMIs” and something has to answer them.
3. **SLR-PLN-03, “select” circled, “provide clearance for ...?”** Left as “select”. The airline’s flight planning system is inside the NAS and is what selects the plan. A clearance is a separate act by ATC, covered by SLR-PLN-06 and SLR-CLR-09.
4. **SLR-PLN-04, “Relevant for weight assumptions?”** Yes. Fuel is part of take-off weight, and the reserve is the floor on how light an aircraft can be in cruise. Kept.
5. **SLR-CLR-08, “An option... leave up to crew w/ response delays?”** Not answered. It reads as an idea for a resolution option in the experiment (the crew decides, after a delay). If you want it, it belongs in `knowledge/models/experiment-options-and-scoring.md`, not here. The requirement is kept as written.
6. **SLR-CLR-09, “Relevance?”** Kept, with the reason in its note: a crew request is today’s only channel for what the aircraft knows about itself. Strike it if you do not want that baseline comparison.

7. **SLR-CLR-13, “What do sources say?”** Section 5.
8. **SLR-EXE-06, “Change to ‘provide no clearance that results in fuel @ dest below threshold’? Change to crew requirement?”** Left unchanged. Claude’s suggestion: the first re-wording says the same thing as SLR-CLR-10 once “can fly” is taken to include arriving with the reserve, so it may not need its own line; the alert itself is a requirement on the flight management system and belongs in the next tier (#TODO item d).
9. **Old open item 4, “Impact of my changes = ?”** The tension between sharing the airline’s data and protecting it is no longer in the requirements, because both sides of the protection (SN-AIR-06 and SLR-INF-05) are struck. Protection is now an assumption about the interface (D-020 item 4, amending D-019 item 3).
10. **Old open items 5 and 6, “Not relevant to D-007”, “Not important.”** The three proposed objective rows (airline data protection, passenger safety, Governance) are not added to `knowledge/models/stakeholder-objective-ontology.md`. SN-GOV-01 keeps tracing to the enterprise objective hierarchy in `knowledge/models/stakeholder-register.md`.

Marks Claude had to interpret. Correct any that are wrong.

11. **SN-AIR-01.** The arrow from “Provide a means for airlines to ... so that ...” points at this row. The sentence was rebuilt on that pattern.
12. **SN-AIR-05.** Read as: ”... into account when a change to that flight’s trajectory is required.”
13. **SN-ENV-01.** “economics ?” was inserted: “as low as safety, economics and capacity allow”. The question mark is yours.
14. **SLR-PLN-02.** “the planned take-off weight” is struck and the word written under it is scribbled out. The sentence now ends “from the performance of the aircraft assigned to the flight”.
15. **SLR-CLR-10.** The scribble covers “at its current weight” and also “Not met today”. The first is removed. The `improved` tag was kept, because the fact has not changed: today the ground system can issue a climb the aircraft cannot make, and finds out when the crew says “unable”. If you meant that the re-worded requirement is met today, change the tag to `baseline` and the verification case goes with it.
16. **SLR-INF-02.** Read as: “The NAS shall give the party that decides a change to an aircraft’s trajectory the information needed to preserve safety and pursue optimality”, with “Implied. Why add separately?” taken to mean that weight, cost index and connection priority should not each have a line. “Pursue optimality” is hard to verify as written; the next tier (#TODO item d) is where it becomes checkable.
17. **SLR-EXE-03.** Read as: ”... from the aircraft’s estimated performance”.
18. **SLR-MEA-01.** Your note is a question (“Change to a/c, its clearance, or the constraints imposed on it?”). Applied as: “record each change to an aircraft’s trajectory, to its clearance, or to the constraints imposed on it”.
19. **SLR-APT-04.** “closed up” became “sealed”, which you wrote with a question mark.
20. **“Turn them into HLRs”** (old open item 2). Read as agreement that requirements with a constituent system as the subject are the next tier down. None were written in this pass; that is #TODO item (d).

Cuts and changes Claude made without a mark on the row.

21. **SN-ATC-05, SLR-FLW-01, -02, -04, -05 and SLR-MEA-04** were struck on the strength of your section note on traffic flow management, not a mark on each row.
22. **SLR-PLN-08** lost “and special use airspace activations”, because the military requirement behind it is struck.
23. **Re-parenting.** Striking a stakeholder requirement left some kept system requirements without a parent. They were moved as follows. To SN-ATC-01 (separation): SLR-PLN-06, SLR-INF-01, SLR-EXE-02 (were under SN-ATC-03) and SLR-CLR-12 (was under SN-ATC-04). To SN-GOV-01 (operating rules): SLR-PLN-04, SLR-PLN-05, SLR-CLR-08, SLR-MEA-01, SLR-EXE-06. To your new SN-CRW-06: SLR-CLR-01, SLR-CLR-02. To SN-CRW-07: SLR-PLN-07, SLR-INF-06, SLR-FLW-03. To SN-CRW-08: SLR-CLR-04.

One consequence to look at. Striking SN-ATC-04 means no stakeholder requirement now says outright that controller workload matters, yet instruction count is one of the experiment’s two scored columns, and workload is one of the five effects D-007 names. The goal SLR-CLR-12 now hangs from separation (SN-ATC-01), on your note that fewer instructions is the best way to keep separation. That is defensible. If a reviewer asks “which stakeholder wants fewer instructions?”, the answer on paper is only SN-ATC-02’s objective rows. Restoring SN-ATC-04 would be the direct fix. Let's bring back SN-ATC-04.

8. Objectives with no requirement → These are derived reqts @ the HLR level.

Rows of `knowledge/models/stakeholder-objective-ontology.md` that no stakeholder requirement picks up, and the reason. The rows are not judged unimportant to the NAS; they are outside the study (section 2).

Category	Objective rows	Why not carried
Airline	Crew cost; Maintenance; Revenue; Dispatch reliability; Flight Safety	The first four are marked Indirect in the ontology’s ‘Trajectory impact’ column. Flight Safety was carried only by the struck SN-PAX-01; separation is carried by SN-ATC-01.
Passenger	all rows	The owner struck all three passenger requirements (“Not D-007”). Connections reach the study through the Airline row ‘Passenger connections’ (SN-AIR-02, SN-AIR-05).
ATC / ANSP	Predictability; Delay (ATC-attributed)	SN-ATC-03 and SN-ATC-05 struck.
Airport	all rows	“All of these are not D-007” (owner).
Flight Crew	Operational flexibility; Schedule/duty constraints	SN-CRW-01, SN-CRW-04 and SN-CRW-05 struck.
Environmental / Societal	NOx; Contrail/climate effects; Noise; Local air quality; Community impacts	Noise: “Not important for cruise” (owner). The others: no source is held, and D-015 notes the altitude link to non-CO2 effects still needs one.
Military	all rows	SN-MIL-01 struck. D-007 makes military operations a boundary actor.

9. How the two example workbooks were used

The owner’s earlier workbooks were written for an airline operations center (AOC) as the system. They were used for their form, not their content.

Carried over from `projects/nas-sos-capstone/prework/Stakeholder Requirements.xlsx`, worksheet "Stakeholder Requirements Groupi":

- Requirements are grouped by the stakeholder who wants them.
- Each is one sentence with the system as the subject and one "shall".
- Broad wishes are allowed at this level ("maximize profits", "minimize delays"); the number comes later.
- Rows that repeat another row are merged, as the workbook did with its "already captured" comments.

Carried over from `projects/nas-sos-capstone/prework/1 System Requirements_10-26_v1.xlsx`, worksheet "Report":

- A compound stakeholder requirement is split into single statements (the workbook split "accurate and up-to-date flight plans" into two).
- A vague word is replaced by something that can be checked (the workbook turned "significant changes in fuel status" into "in excess of 5% of expected fuel burned").
- A rule is named by its paragraph where one is known (the workbook named the CFR part for crew fatigue).
- There are more system requirements than stakeholder requirements.

Not carried over:

- The content. Most workbook rows say what the AOC owes another party. Here that other party is usually inside the system, so the row becomes an exchange between two constituent systems.
- Rows about how an airline runs itself (budget, corporate values, IT approval, cybersecurity training, record keeping). They do not touch the trajectory-intent chain.
- Numbers. The workbook's 60 minutes for filing a flight plan and 5 percent fuel deviation are not reused, because no source held in this repository gives them.

How the workbook's stakeholders map onto this project's. Each row states the abstraction and the reason, so it can be defended or changed.

Workbook stakeholder	This project's group	Abstraction	Why
Air Traffic Control (ATC)	Air traffic control and traffic flow management (FAA Air Traffic Organization)	identical	Same stakeholder. The direction reverses: the workbook asked what the AOC owes ATC; here ATC is inside the system, so the rows say what ATC wants from the whole.
Customers	Passengers	renamed	D-007 uses Passengers and treats them as a boundary actor.
Pilots & Flight Crew	Flight crew (captain and first officer)	identical	Cabin crew are left out: no authority over in-scope decisions (nas_context_diagram.sysml, cabinCrew comment).
FAA, EASA & all Aviation Authorities	Regulator (FAA as rule maker; Department of Transportation)	many to one	D-007 scopes to U.S. domestic Part 121, so only the FAA and DOT rules apply.
Airports	Airport operator	identical	
Airline Fleet Planning; Airline Management; Airline & 3rd Party Maintenance & Engineering; Airline Safety & Security Department; Airline IT	Airline (Part 121 scheduled passenger operator)	many to one	In the workbook the AOC was the system, so the rest of the airline stood outside it as separate stakeholders. Here the whole airline is one constituent system (Flight Operations), so its departments are one stakeholder. Their objectives survive as rows in the Airline table of knowledge/models/stakeholder-objective-ontology.md.
Pilot Unions	Flight crew (captain and first officer)	many to one	The union's interest in the workbook was duty and rest limits. The requirement that carried it (SN-CRW-05) was struck in the hand review as outside the study.
(none)	Communities and environmental organizations	gap in the workbook	Added from knowledge/models/stakeholder-objective-ontology.md and knowledge/models/stakeholder-register.md.
(none)	Military airspace users	gap in the workbook	Added from D-007 (boundary actor) and the suaToTfm link in nas_context_diagram.sysml. Its one requirement was struck in the hand review.

Evidence grades used in this file follow the review key in `projects/nas-sos-capstone/cameo_models/nas_context_diagram.sysml`:

- [A] rule or FAA order held in the source register
- [B] published study held in the source register
- [SME] owner's industry knowledge, to be confirmed by the owner
- [C] general domain knowledge, not yet tied to a source in this repository

Names in `code font` in the Basis column are keys in `evidence/sources/references.bib`. A grade says where the idea comes from. It does not mean the cited paragraph was re-read in the session that wrote this.

10. Coverage check

Run when these files were regenerated on 2026-10-10, after the hand review:

- Stakeholder requirements with no system requirement: 0 of 12.
- System requirements with no stakeholder parent: 0 of 28.
- System requirements with no allocation: 0 of 28.
- System requirements with no "In the study" note: 0 of 28.
- `syside check` over the whole `projects/nas-sos-capstone/cameo_models/` folder: "All checks passed!"

The table of what became of the ten placeholder needs and six placeholder requirements of 2026-09-19 (old section 8) was dropped from this file; it is on page 13 of the marked-up PDF.